

| Applied Data Analytics - Answer Sheet |                             |                                           |
|---------------------------------------|-----------------------------|-------------------------------------------|
| Username                              | Attempt start time          | Attempt end time                          |
| U15FY25S0049                          | Tue Jun 30 2026 15:50:26 PM | Tue Jun 30 2026 16:38:26 PM               |
| Total Questions (Includes Optional)   | Attempted Questions         | Unattempted Questions (Includes Optional) |
| 55                                    | 35                          | 20                                        |

| Question No. | User Answer                                                       |
|--------------|-------------------------------------------------------------------|
| Q. 1.        | Option B.<br>import pandas as pd                                  |
| Q. 2.        | Option B.<br>Comma-Separated Values                               |
| Q. 3.        | Option A.<br>plt.show()                                           |
| Q. 4.        | Option B.<br>Generates statistical summary of numerical columns   |
| Q. 5.        | Option C.<br>pd.read_csv('file.csv')                              |
| Q. 6.        | Option C.<br>df['Sales'].mean()                                   |
| Q. 7.        | Option B.<br>df.isnull().sum()                                    |
| Q. 8.        | Option C.<br>Returns a DataFrame with rows containing NaN removed |
| Q. 9.        | Option C.<br>plt.title()                                          |
| Q. 10.       | Option A.<br>df.types()                                           |

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| Q. 11. | Option C.<br>Pie chart                                                     |
| Q. 12. | Option B.<br>Replaces all NaN values with 0                                |
| Q. 13. | Option B.<br>Frequency distribution of a single variable                   |
| Q. 14. | Option C.<br><code>np.zeros((3,4))</code>                                  |
| Q. 15. | Option C.<br><code>df.drop_duplicates()</code>                             |
| Q. 16. | Option B.<br>A sequence of data processing steps from collection to output |
| Q. 17. | Option B.<br><code>df.sort_values('col', ascending=False)</code>           |
| Q. 18. | Option C.<br>Shows relationship between two numerical variables            |
| Q. 19. | Option B.<br>Patterns and changes in data over time                        |
| Q. 20. | Option B.<br><code>df.rename(columns={'old':'new'})</code>                 |
| Q. 21. | Option A.<br><code>df.column('Age')</code>                                 |
| Q. 22. | Option B.<br>Using analyzed data results to guide business decisions       |
| Q. 23. | Option D.<br>Data encryption                                               |
| Q. 24. | Option C.<br>Groups data by City and computes mean of numerical columns    |
| Q. 25. | Option B.<br>Converting raw data into a format suitable for analysis       |

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|--------|---------------------------------------------------------------------------------------|
| Q. 26. | Option B.<br>df.groupby('Year')['Sales'].mean().idxmax()                              |
| Q. 27. | Option C.<br>Median                                                                   |
| Q. 28. | Option C.<br>Investigate why 45% is missing then impute with median or domain logic   |
| Q. 29. | Option B.<br>Line chart, Pie chart, Bar chart                                         |
| Q. 30. | Option B.<br>Whether the removed rows contained disproportionately high-value records |
| Q. 31. | -                                                                                     |
| Q. 32. | -                                                                                     |
| Q. 33. | -                                                                                     |
| Q. 34. | -                                                                                     |
| Q. 35. | -                                                                                     |
| Q. 36. | -                                                                                     |
| Q. 37. | -                                                                                     |
| Q. 38. | -                                                                                     |
| Q. 39. | -                                                                                     |
| Q. 40. | -                                                                                     |
| Q. 41. | -                                                                                     |
| Q. 42. | -                                                                                     |
| Q. 43. | -                                                                                     |
| Q. 44. | -                                                                                     |
| Q. 45. | -                                                                                     |
| Q. 46. | -                                                                                     |

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| Q. 47. | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q. 48. | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q. 49. | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q. 50. | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q. 51. | (1)identify the missing values :dicount_%has 12% missing values and customer_age_group has 5% missing values (2) handel disount_% since :it is a numerical column fill the missing values using the median (3) handel customer_age_ group: categorical column,fill the missing values using the mode (4) do not remove rows;avoid deleting rows because only small percentage age of data is missingb (5) validate the data : after filling the missing values check the data set to ensure there are no missing values left |
| Q. 52. | 1. create reveneu column using the formula $revenue = units\_sold * unit\_price * (1 - dicount\_%/100)$ 2. group the data by city by using <code>groupby (city)</code> 3. calculate the total revenue for each city using <code>.sum()</code> 4. find the city with the highest revenue using <code>.idxmax()</code> 5. display the result                                                                                                                                                                                   |
| Q. 53. | 1. use a grouped bar chart to compare product categories 2.the chart clearly shows which product categorie has the highest sales in each age group 3.compare sales of nall product categories for every customer age group 4.identify the best selling product category foe each age group 5.these help the company understand customer reference for make better marketting                                                                                                                                                 |
| Q. 54. | (a)1.use a line chart 2.show monthly sales trends (b)3.increase stock in high sales months 4.offere discounts in low sales months improve sales planning and inventory                                                                                                                                                                                                                                                                                                                                                       |
| Q. 55. | 1.present key sales insight with simple charts and graphs 2.highlight top products ,cities,trends 3.use simple language for non technical stakeholders 4. provide clear business recommendation based on the analysis 5. focus on action able insights to improve sales and descion making                                                                                                                                                                                                                                   |